

The same use case, applied elsewhere

	 Car Park Lots Availability	 Hotdesking Facility Management	 Aerospace MRO Manpower Planning
Use case	A person sees carpark occupancy rate and forecast and decide whether to drive or take public transport.	A facility manager sees the facility occupancy data to monitor, analyze and optimize facility operations.	An operation manager sees the available technicians and task orders to forecast and manage manpower planning.
DATA	Objective & non-personal - Live lot counts from LTA/HDB APIs - Weather conditions	Becomes more complex - Real-time occupancy analytic - User's attributes: proximity to team, preferred amenities, personal schedule and historical occupancy patterns - IoT sensors, building and facility management system	Highly complex and dynamic - Shifts, leave and overtime schedule - Skills, experience, certification and licenses - Task requirements varied by aircraft/component type and regulated by aviation authority (e.g. Civil Aviation Authority Singapore)
RISK	- Driver frustration - Traffic congestion - Parking space overutilized or underutilized	- Privacy – user's location - Prioritize space optimization over user experience during unexpected surge in demand - Space may be unnecessarily reconfigured, resulting in additional operating cost	- Regulatory and safety risk - Over-optimization for productivity and underestimated workload - E.g.,: A fatigued technician or trainee deployed for a task touches airworthiness and labour concerns
OVERSIGHT	- Can the tool handle increasing user load? - Is the occupancy rate updated? - Is the prediction based on weather condition accurate?	- Are there any "ghost" occupancy? - Who can access user information? - Is approval required for space re-layout?	- Is the technician assigned to the work qualified and sufficiently trained to do the work? - Is the workload evenly distributed? - Is the estimation of task duration accurate? (e.g., defect discovered during inspection)

Conclusion: The capacity-sensing concept generalizes well across industries. However, prediction and recommendation require robust AI models that can account for data complexity, risk, uncertainty and level of human oversight required. So, we would not recommend the use case for MRO manpower planning.